work and you're a hundred million miles from the nearest dentist? Or could it be done by an endodontist on Earth, using telepresence?

Wernher von Braun was the Nazi-American engineer who, more than anyone else, actually took us into space. His 1952 book *Das Marsprojekt* envisioned a first mission with 10 interplanetary spacecraft, 70 crew members, and 3 "landing boats." Redundancy was uppermost in his mind. The logistical requirements, he wrote, "are no greater than those for a minor military operation extending over a limited theater of war." He meant to "explode once and for all the theory of the solitary space rocket and its little band of bold interplanetary adventurers," and appealed to Columbus' three ships without which "history tends to prove that he might never have returned to Spanish shores." Modern Mars mission designs have ignored this advice. They are much less ambitious than von Braun's, typically calling for one or two spacecraft crewed by three to eight astronauts, with another robotic cargo ship or two. The solitary rocket and the little band of adventurers are still with us.

Other uncertainties affecting mission design and cost include whether you pre-emplace supplies from Earth and launch humans to Mars only after the supplies are safely landed; whether you can use Martian materials to generate oxygen to breathe, water to drink, and rocket propellants to get home; whether you land using the thin Martian atmosphere for aerobraking; the degree of redundancy in equipment thought prudent; the extent to which you use closed ecological systems or just depend on the food, water, and waste disposal facilities you've brought from Earth; the design of roving vehicles for the crew to explore the Martian landscape; and how much equipment you're willing to carry to test our ability to live off the land in later voyages.

Until such questions are decided, it's absurd to accept any figure for the cost of the program. On the other hand, it was equally clear that SEI would be extremely expensive. For all these reasons, the program was a nonstarter. It was stillborn. There was no effective attempt by the Bush Administration to spend political capital to get SEI going.